Saudi Arabia Domestic Tourism Analysis & Visualization (2018-2023)

1-Introduction:

This dataset contains monthly records describing domestic tourism activity in Saudi Arabia from 2018 to 2023. It was obtained from the Kaggle platform and includes variables such as the number of trips, tourism spending, nights stayed, destination and origin provinces, purpose of travel, and average temperatures for both origin and destination regions.

The dataset was selected because it relates to a key sector undergoing major growth in Saudi Arabia and provides a rich mix of temporal, geographical, quantitative, and categorical attributes, making it suitable for multi-dimensional visual analysis.

Two preprocessing steps were applied to prepare the data:

1.The dataset was checked for duplicated rows, and none were found.

2.Missing temperature values were handled using hierarchical mean imputation based on (region + month), then region alone, and finallythe overall mean, ensuring pattern preservation without removing any records.

2-Data Abstraction:

1) IdentifyAttributesandDataTypes

Quantitative Attributes:

- •trips
- •spendSAR
- •nights
- •origin_temp
- •destination_temp

Geographical Attributes:

- •originProvinceNameEn
- •destinationProvinceNameEn

Categorical Attributes:

- •visitPurposeEn

Temporal Attributes:

- •year
- •month

Derived Attributes (Calculated Fields):

- •trip volume
- •avg nights per Trip
- •Season by Month
- •Avg_Destination_Temp

2) Data Relationships

• Items:

Each row represents a single monthly tourism movement record between an origin and a destination province.

• Fields / Attributes:

The columns describe each record and include quantitative, categorical, temporal, and geographical variables.

• Links:

Logical relationships exist between variables, such as:

•the flow from origin province to destination province (origin → destination),

•the relation between temperature and trips,

•the relation between travel purpose and spending or trip volume.

• Hierarchies:

•Temporal hierarchy: Year → Month

•Seasonal hierarchy (derived): Season → Months

•Data Structure:
The dataset is a flat table with no multi-level structure, but it supports aggregation and grouping by time or region.

3-Task Abstraction:

1. LineChart

The main analytical task supported by this line chart is analysing change over time, as it shows how the total number of domestic trips changes annually from 2018 to 2023.
By observing the trend across years, the user can identify periods of increase, decrease, or fluctuations.
This directly relates to the project's research question about the factors influencing domestic tourism volume, as the chart reveals that time (year) is an important factor affecting the number of trips during the study period.

2. Tree map

The primary analytical task supported by this visualization is identifying seasonal patterns in the number of domestic trips by comparing the four seasons within each of the top three regions. This helps highlight peak seasons and low-activity periods.
This task directly supports the project's main research question about the factors influencing domestic tourism volume, as the tree map shows that season (time of year) is an important factor affecting the number of trips during the study period.

3. Bubble Scatter Plot

Correlate
The scatter plot helps identify the **relationship** between *Spend SAR* and *Average Nights per Trip*, revealing whether higher spending is associated with longer or shorter stays.

Relation to the Research Question:
This helps answer the project's main question about the factors influencing domestic tourism volume, as variations in spending and stay duration explain how different tourist behaviors contribute to overall trip patterns.

4. Choropleth Map

This visualization supports the task of identifying spatial patterns by comparing regions based on their total number of trips, while using color to represent the average destination temperature. This helps highlight regional differences and shows which areas receive higher or lower tourism activity.

This analysis relates to the project's main question about the factors influencing domestic tourism volume. The map indicates that geographic location and climate may play a role, as some warmer regions record higher numbers of trips compared to cooler northern regions.

5. BarChart

This chart supports the task of comparing and ranking categories, showing how domestic trips differ by visit purpose (Leisure, VFR, Religious, Business, Other). By observing bar heights and percentages, the user can identify which purposes dominate and which are less significant.
This analysis relates to the project question: What factors influence the volume of domestic tourism in Saudi Arabia from 2018 to 2023? The chart demonstrates that visit purpose is a key factor, with leisure and social visits emerging as the main drivers of tourism activity.

4-Design Justification:

1-Line Chart

Thisvisualization shows the total number of domestic trips for each year from 2018 to 2023. A line chart was chosen because it is the most effective way to display temporal data and reveal overall trends over time.
Visual Encodings:
•Position: Years are placed on the horizontal axis and total trips on the vertical axis, making it easy to compare values across years.
•Color: A single color is used since the chart represents one overall trend, not multiple categories.
•Shape: A continuous line connects yearly values to clearly show the progression and change over time.
Support for the Analytical Task:
This design directly supports the task of analysing change over time, as it clearly highlights increases, decreases, and fluctuations in domestic trip volume. It helps reveal how time functions as a key factor influencing tourism levels during the study period.
Alternative Options:
Bar charts, area charts, and scatter plots were considered but rejected because they do not show continuous temporal trends as clearly as a line chart.

2-Tree map

This tree map displays the top three regions based on the total number of domestic trips and shows the four seasons within each region as rectangles that vary in size and color according to the total number of trips in each season.
The tree map was chosen because it is well-suited for representing hierarchical data (Region → Season) and clearly highlights seasonal differences without visual clutter.
Calculated Field:
A calculated field (Season by Month) was created to convert month numbers into seasons (Winter, Spring, Summer, Autumn). This step was necessary to enable seasonal analysis, and it is considered data preparation rather than visual encoding.
Visual Encodings:
•Size: The size of each rectangle represents the total number of trips in that season; larger rectangles indicate higher activity.
•Color: A gradient is used to show activity levels, with darker colors indicating higher trip counts.
•Position: Regions and seasons are ordered from highest to lowest total number of trips, making the most active seasons appear first.
•Labels: Region and season names were added for clarity; these labels are not visual encodings.
•Filter: Only the top three regions were displayed to reduce clutter and maintain focus.
How It Supports the Task:
The design supports the task of identifying seasonal patterns by clearly showing which seasons have high or low trip volumes within each region. This highlights the influence of seasonality on the total number of trips, directly supporting the project's main research question.
Alternatives:
Other charts, such as bar charts or line charts, were considered but rejected because they cannot represent the hierarchical structure or seasonal proportions as effectively as a tree map.

3-Bubble Scatter Plot

Whythis visualization?

A **Bubble ScatterPlot** was chosen because it effectively shows:

- The **relationship** between *Spend SAR* (quantitative) and *Avg Nights per Trip* (quantitative),
- The **distribution of visit purposes* through color,
- The **trip volume impacts** bubble size.

This makes it ideal for multi-variable analysis.

Visual Encoding Justification
Position (X & Y):

- X-axis: **Spend SAR**
- Y-axis: **Average Nights per Trip**
  Position isthemost accuratevisual channel, making it suitable for comparing two quantitative variables.

Color:
Color encodes **Visit Purpose En**, which is perfect for distinguishing categories:

- Leisure (Orange)
- Religious (Green)
- VFR (Blue)

This supports easy pattern recognition and grouping.
Size:
Bubble size represents **Trip Volume**.
Size encoding is effective for showing magnitude differences without taking extra space.
Labels:
Destination names were added to allow easier interpretation during analysis without needing tooltips.

How it supports the tasks:

- **Correlation:** Theprimary analytical task supported by this bubble scatter plot is identifying the relationship between spending (Spend SAR) and average nights per trip, while distinguishing visit purposes.
  By examining how spending changes relative to stay duration across different purposes, the visualization reveals behavioral differences between regions and visit types.
  overall trip patterns

Alternative Designs Considered:

- A normal scatter plot was considered, but it cannot represent trip volume effectively.
- A bar chart was rejected because it cannot show relationships between two quantitative measures.
- A heatmap was rejected because it does not communicate correlations clearly

---

4-Geographic Map

What it shows and why it was chosen:
This map displays the regions of Saudi Arabia, using color to represent the average destination temperature, while showing the total number of domestic trips as labels. A choropleth map was chosen because it is the most appropriate visualization for geographic data and allows clear spatial comparison between regions.

Visual Encodings:
•Position: The map uses the actual geographic coordinates (latitude and longitude), allowing users to directly compare regions based on their real spatial locations.
•Colormap colour gradient was applied to represent Avg_Destination_Temp.
Darker shades indicate higher temperatures, while lighter shades indicate cooler regions.
This highlights climate differences across regions.
•Shape: Each province is represented using its true geographic polygon, helping users quickly recognize regions and interpret spatial patterns.
•Labels: Each region displays its name and SUM (trips) to make the map readable without relying on tooltips.
•Calculated Field: The visualization uses the calculated field Avg_Destination_Temp to accurately show the average temperature for each region.
How it supports the defined task:
This design supports the task of identifying spatial patterns, as it allows users to compare regions based on both trip volume and temperature, revealing how geographic and climate differences might influence domestic tourism levels.
Alternative designs considered:
Bar charts or line charts were not used because they cannot represent spatial location, whereas the map provides clearer geographic insights.

5-Bar Chart

This chart shows the distribution of domestic trips by visit purpose (Leisure, VFR, Religious, Business, Other). A bar chart was chosen because it is suitable for categorical data, making it easy to compare values and identify which purposes dominate.

- Position: Visit purposes on the x-axis, trip numbers on the y-axis, showing differences clearly.
- Color: A single color was used to keep focus on values.
- Shape: Bars represent each category, with height encoding trip numbers and percentages for accuracy.

This design supports comparison and ranking tasks, highlighting the impact of visit purpose as a key factor influencing domestic tourism in Saudi Arabia.

5-Story and Insights

The line chart shows an increase in the volume of trips after 2020, prompting an analysis of when this growth occurs within the year. The tree map revealed that spring and summer are the peak periods, which was supported by the bar chart showing that leisure is the main motivation for travel. The scatter plot then explained the differences in travel patterns across regions by examining the relationship between length of stay and expenditure by purpose, highlighting a distinctive tourism identity for each area. Finally, the map confirmed that these variations are linked to spatial and climatic factors affecting demand across regions. Together, these visualizations provide a comprehensive view of the factors influencing domestic tourism.